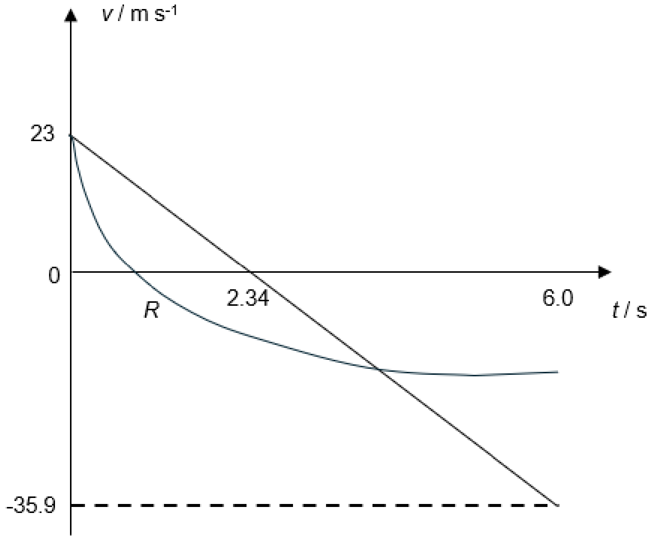
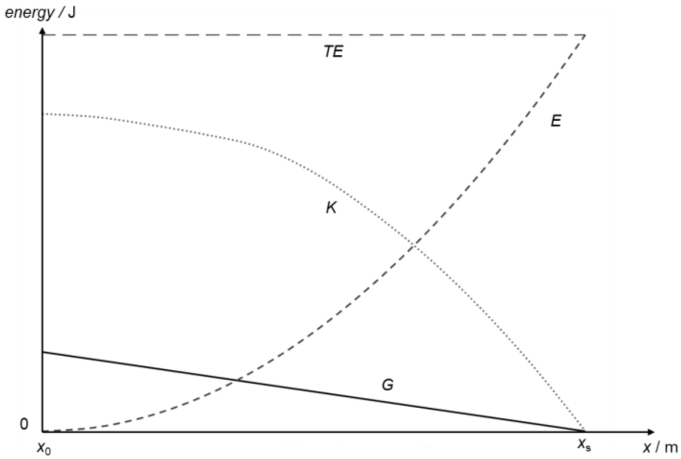
2	
(a)(i)	Taking upwards as positive, $v^2 = u^2 + 2as$ $0^2 = u^2 + 2(-9.81)(27)$
	$u = 23.02$ $= 23$ m s^{-1} (2 s.f.) (shown)
	

(a)(ii)	Straight line with negative gradient.
	Initial velocity of 23 m s^{-1}. Max height at 2.34 s. Graph stops at 6.0 s with velocity of -35.9 m s^{-1}. $\frac{v-u}{t}=g$ $t_{\text{max height}}=\frac{v-u}{g}=\frac{0-(-23)}{9.81}=2.34\text{ s}$ At $t=$ 6.0 s $v=u+at=23-(9.81)(6.0)$ $=-35.9\text{ m s}^{-1}$
(a)(iii)	Steeper slope before v = 0. Gentler slope after v = 0.
	Gradient at v = 0 should be parallel to graph in **(a)(ii)**.
(b)	
	GPE linear with negative gradient.
	EPE parabolic shape starting from x_0.
	KE shape and correct KE value at x_0 and x_s (TE is constant).
3	